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GrayCo Odor Control

Employee Training Manual

GrayCo's odor control training: smoke, pets, mold, trauma scenes, tobacco and more. Read every section, then answer the review questions at the end. In the CRM, open Training to tick off each part and take the quiz (pass = 16 of 20). It covers what the IICRC Odor Control Technician (OCT) class covers, so it is also your study guide for that exam.

1. How we smell

An odor is a gas that moves from a source into the air and is picked up by the **nose**. Your nose is your best tool for finding an odor and judging how strong it is.

- The brain identifies odors in the **olfactory lobe**. When odor molecules touch the nerve endings in your nose and the signal reaches the brain, that is **real odor**.
- **Heightened awareness odor** (psychological odor) is an odor people believe they smell because of experience, suggestion or memory, like the family that still smells smoke after the house is clean.
- To be smelled, a molecule has to be **volatile** (it evaporates), at least slightly **water soluble**, and soluble in fats.
- People get **acclimated** to odors and stop noticing them, like the baker who no longer smells bread. A fresh nose is a better judge.
- Odor is strongest when **temperature and relative humidity are high**.
- **Adsorption** is odor molecules holding onto the surface of a material. Porous materials such as fabric, wood, paper, cork, flat paint and unsealed concrete are **natural wicks**.

2. The four principles of odor control

Every odor job follows the same four steps:

1. **Remove the source**, as much of it as you can. This is always where deodorizing starts.
2. **Clean the source area.**
3. **Penetrate**: get the odor counteractant into every surface and space the odor reached.
4. **Seal**: a **sealer** locks in odor molecules so they can't escape. Sealers are rated by their **perm** (permeance) rating; the closer to zero, the better they hold odor in.

Skipping step 1 is the biggest mistake in the business. If odor comes back, go find the source.

3. How deodorizers work

- **Air exchange**: push bad air out and pull fresh air in with **exhaust fans and air scrubbers**.
- **Adsorbents**: activated charcoal (carbon) and baking soda trap odor molecules on their surface.

- **Neutralization**: balances a chemical odor, such as baking soda on an acid spill.
- **Oxidation**: adds **oxygen** to break down organic material. Hydrogen peroxide, chlorine bleach and ozone are **oxidizers**.
- **Biocides**: kill or slow down the microorganisms that cause odor.
- **Counteraction**: most smoke deodorants combine **masking** and **pairing** agents. A **pairing agent** combines with the bad odor to form a new compound with no odor.
- **Masking**: covers a bad odor with a stronger, nicer one. Time-release granules, oils and gels work on the **time-release** principle.
- **Bioenzymatic digestion**: enzymes made by special bacteria eat organic material such as urine, milk, eggs, fish and spilled fuel oil.

4. Equipment

A **micron** (one thousandth of a millimeter) is how we measure **droplet and particle size**.

- **Pressure / compression sprayers**: big droplets (hundreds of microns). The most control; best when you want to coat a surface.
- **Wet foggers / misters**: 25–100 micron droplets. Can over-wet and stain **moisture-sensitive** materials if the operator isn't careful.
- **ULV foggers**: cold-fog **water**- or alcohol-based deodorants at **8–15 microns**. Droplets stay in the air 5–6 hours.
- **Thermal foggers**: heat petroleum-based deodorant into **0.5–2 micron** droplets that act like smoke and modify the odor (a **masking** treatment).
- **Ozone (O3)**: an oxidizing gas made on site by an **ozone generator**. It is **heavier than air**, so use a fan to move it. It harms plants and natural (latex) rubber. Too much causes coughing, sneezing, nosebleeds and eye irritation. Nobody stays in the space while it runs; it takes 24–72 hours, and most of it is gone 10–15 minutes after you turn it off. Ventilate before anyone goes back in.
- **Hydroxyl generators**: UV light plus water vapor makes hydroxyls that oxidize odor.
- **Exhaust fans and air scrubbers** dilute or remove odors and toxic gases; carbon filters adsorb odor.

5. Smoke odors

- **Protein** odors (burned meat or poultry) are persistent; yellow-brown greasy residue, usually needs wet cleaning.
- **Natural** odors (wood, paper, cotton): gray or black, dry and powdery.
- **Synthetic** odors (plastics, synthetics): black residue that smudges.
- Smoke gets in by **heat, pressure, impingement** (splatter), **magnetism** (drawn to metal) and **ionization** (smoke webs). Driven (hot) smoke marks vertical surfaces; free-floating (cold) smoke settles on horizontal ones.
- **Higher temperatures greatly increase** how deep odor goes into porous materials.

The three-step smoke odor process

Use **multiple techniques**; one method alone never finishes the job. Odor work should **precede** normal cleaning and repairs.

1. **Suppression spray**: water-based counteractant on every surface in the source area (preliminary detoxification).

2. **Time-released counteractant** granules on horizontal surfaces, attics, crawl spaces and plenums.
3. **Thermal fogging**, which follows the same paths the smoke took.

Fogging safety: evacuate unprotected **persons, pets** and plants. Wear respiratory, **eye** and skin protection. Tell the **fire department** first. Turn off pilot lights, because over-fogging a tight space risks **explosion**. Too much fog leaves a slippery **oily** residue. Cover the smoke detectors, then uncover them afterward.

6. Microbial odors

- **Pathogenic** means disease-causing.
- **Bacteria** are **single-celled** organisms with no chlorophyll. They split in two as often as every 15–30 minutes.
- **Fungi** (molds, yeasts, mushrooms) feed on living or dead material. Indoors they like **68°–86°F**, the temperatures people like. They need **moisture** to germinate from **spores**.
- Mold odor is **MVOCs**: microbial volatile organic compounds.
- **Viruses** are parasites that only live and reproduce inside the cells of a living host.
- In the U.S. the **EPA** regulates disinfectants and antimicrobials; in Canada it is **Health Canada**.
- **Biocides** (antimicrobials) are the tool for odors caused by microorganisms. Keep the surface wet for the label's dwell time. **Following the label directions is the law.**

Read the label

- **Sterilant**: kills microorganisms and their spores.
- **Disinfectant**: kills most microorganisms, but **not necessarily spores**.
- **Sanitizer**: reduces microorganisms to a level that is **safe for human use**.
- **-cide** (bactericide, fungicide) **kills**. **-stat** (bacteriostat, fungistat) **limits** growth.
- Signal words: **Danger** = highly toxic, **Warning** = moderately toxic, **Caution** = low toxicity.

7. Pet and animal odors

- If the pet is still living there, **never guarantee** urine odor removal; it will go back to the same spot.
- Urine-soaked **carpet pad is usually discarded**. Severe odor may need a **bio-enzymatic** (digesting) deodorant.
- Find out how many animals there are, the **degree of penetration**, and how much urine there is. Urine soaks down through the **multiple layers** of carpet, pad and subfloor.
- An **ultraviolet (black) light** is the best way to find urine. Moisture meters and bright lights help too. Male dogs mark along walls; females squat in the open.
- Fresh urine is **acidic**; as it dries it turns **alkaline**. The dried **urine salts** are hygroscopic: they pull moisture from the air and release odor again. An **acid rinse** neutralizes and dissolves those salts.
- Options, from cheapest to best: customer-applied products → **spot treatment** (inject, acid rinse, flush with clear water, enzyme, dwell, **thorough extraction**) → **total decontamination**.
- Concrete is the easiest floor to clean (acid cleaner, then a solvent-based sealer). Vinyl tile and linoleum are the hardest and usually come out.
- Seal odor-damaged wood and walls with a **nonporous sealing compound** (pigmented shellac, lacquer or alkyd), never latex.

- **Dead animals**: the hard part is **finding the carcass** (borescope, cadaver dogs, wetting and heating to bring the odor out).
- **Skunk and decomposition** odors respond to **oxidation**. **Chlorine bleach** (1:10) works on durable **colorfast** surfaces such as concrete, but it discolors nylon, corrodes metal and destroys protein fibers. **Hydrogen peroxide** works on colorfast surfaces too.

8. Trauma scenes and bloodborne pathogens

Death and blood cleanups bring two main bloodborne risks: **Hepatitis B (HBV)**, which attacks the liver, and **HIV**, which can lead to **AIDS**. The odor comes from decomposing **proteins**.

- A **bloodborne pathogen** is a harmful microorganism in blood, body fluids or other potentially infectious materials (OPIM). **Saliva and blood** can both carry them.
- The program starts with a written **exposure control plan**: how a worker could be exposed and how we prevent it.
- **PPE is mandatory** whenever there is a risk of **exposure** to urine, feces, blood or sewage.
- **Universal precautions**: treat **all** blood and body fluids as potentially **infectious**.
- Soaked fabrics go in labeled containers and are handled as **biohazard** waste. Biohazard labels are printed on **fluorescent red-orange**.
- GrayCo offers the **Hepatitis B vaccine** free to anyone who may be exposed. If someone who isn't vaccinated is exposed, the first shot must be within **24 hours**.
- **Exposure incident**: splash in the eye → flush with clean running water for **15 minutes** and see a doctor. On the skin → wash with **soap** and water. GrayCo must **document** the route and circumstances of the exposure, so tell your supervisor right away.
- Finish by cleaning every salvageable surface with a proper **disinfectant**, then seal porous surfaces (not with latex paint).

9. Tobacco, HVAC, new carpet and leftover odors

Tobacco (ETS)

- Tobacco odor comes from sticky residue built up on surfaces, especially **porous surfaces**, ceilings and car headliners.
- The particles are **0.01 to 1.0 microns**, so they get everywhere, including the HVAC.
- The key step is **washing the affected surfaces**. Then run the three-step process: suppression spray, counteractant with an absorbent, thermal fog. Seal with pigmented shellac where needed.

HVAC

- **Never fog or spray liquid deodorizer into a forced-air system.** If someone becomes sensitive to it, the only fix is replacing the ductwork.
- Clean the components and change the filters. Time-release counteractant or carbon can go in the plenum.

New carpet

- New carpet smell is 4-PCH gassing off the latex backing. The first step is **ventilation** (air exchange). A "bake-out" with heat and an adsorbent can help.

When the odor comes back

- The biggest mistake in odor work is failing to find and remove **the source**. Smoke odor does not go away on its own. Sniff-test baseboards, outlets and cavities, and go back to principle 1.

Review questions

Circle the best answer.

1. The technician's best tool for finding an odor and judging how strong it is:
- A. A moisture meter
- B. The nose
- C. A UV light
- D. An air sampler

2. Odors people believe they smell because of memory or suggestion are called:
- A. Real odor
- B. Heightened awareness odor
- C. MVOCs
- D. Complex odor

3. Getting used to an odor until you no longer notice it is:
- A. Adsorption
- B. Acclimation
- C. Oxidation
- D. Penetration

4. Deodorization almost always begins with:
- A. Thermal fogging
- B. Sealing
- C. Removing the source
- D. Ozone

5. Which principle of odor control is "Penetration"?
- A. First
- B. Second
- C. Third
- D. Fourth

6. A product that combines with a malodor to form a new odorless compound is a:
- A. Masking agent
- B. Pairing agent
- C. Biocide
- D. Adsorbent

7. ULV foggers make droplets of about:
- A. 0.5–2 microns
- B. 8–15 microns
- C. 25–100 microns
- D. Hundreds of microns

8. Thermal foggers make droplets of about:
A. 0.5–2 microns
B. 8–15 microns
C. 25–100 microns
D. 1/8 inch

9. Ozone gas is:
A. Lighter than air
B. Heavier than air
C. The same weight as air
D. Not a gas

10. Ozone can damage:
A. Glass
B. Plants and natural (latex) rubber
C. Concrete
D. Steel studs

11. The most effective system for intense smoke odor is:
A. One strong product
B. Multiple techniques
C. Ozone only
D. Painting over it

12. Step 1 of the three-step smoke odor process is:
A. Thermal fogging
B. Time-release granules
C. Suppression spray
D. Sealing

13. The most serious hazard of over-fogging petroleum deodorant in a tight space is:
A. Staining
B. Possible explosion
C. Mold growth
D. Noise

14. Indoor fungi grow best between:
A. 32°–50°F
B. 50°–60°F
C. 68°–86°F
D. 100°–120°F

15. A product with the suffix "-stat" on the label:
A. Kills organisms
B. Limits or controls growth
C. Is a sterilant
D. Is a deodorant only

16. Reducing microorganisms to a level safe for human use is called:

A. Sterilizing
B. Disinfecting
C. Sanitizing
D. Masking

17. If the pet will keep living in the home, you should:

A. Guarantee the work for 1 year
B. Never guarantee urine odor removal
C. Only use ozone
D. Refuse the job

18. As urine dries it changes from:

A. Alkaline to acid
B. Acid to alkaline
C. Neutral to acid
D. It does not change

19. Treating all blood and body fluids as potentially infectious is called:

A. Universal precautions
B. Exposure control
C. Sanitizing
D. Decontamination

20. An unvaccinated tech exposed to blood must get the first Hepatitis B shot within:

A. 24 hours
B. 3 days
C. 1 week
D. 30 days

Employee name: _____ Date: _____

Score: _____ of 20 Reviewed by: _____